

The called strike zone shrank twice: 2025 and 2026 against 2022-2024

Introduction

The called strike zone shrank in 2026, when ABS challenges began. Umpire grading had already tightened in 2025, from a buffer two inches outside the edge. Published estimates start from 2025 or fold it into a trend (Lee et al., arXiv, 22 September 2026), so neither step has an interval against old-rule seasons. This paper measures both steps against 2022-2024 and argues that more of the contraction came in 2026, although the zone was already smaller before challenges began.

Methods

To put five seasons on one zone, I re-projected each 2022-2025 pitch to the plate midpoint Statcast adopted in 2026. I fitted called-strike probability to 1,830,231 called pitches from 12,060 regular-season games and read each season's zone as its 50 percent contour. It agreed with the ABS verdict on 10,164 of 10,168 challenged pitches. I measured both steps net of the 2022-2024 trend, pooled umpires, and pre-registered a placebo between 2023 and 2024, two seasons under one rule.

Results

The placebo failed: with no rule change, the zone grew 10.8 (90% CI 7.7 to 13.9) square inches, so every step below is descriptive. As Figure 1 shows, the zone measured 494.6, 484.3 and 495.1 square inches from 2022 to 2024, then 471.9 and 439.1, a loss of 56.0 (95% CI 52.0 to 59.6) since 2024. The 2025 step was 23.4 (95% CI 18.6 to 28.3) square inches and the 2026 step 33.1 (95% CI 28.7 to 37.5). As Table 1 shows, the 2025 step came mostly from the sides, in by 0.41 (95% CI 0.33 to 0.49) inches each. In 2026 the top came down 0.63 (95% CI 0.50 to 0.76) inches, an interval that reads slightly too narrow, and the bottom rose 0.65 (95% CI 0.55 to 0.74). On the published convention, 2025 at the plate front and 2026 at mid-plate, the contraction is 20.0 (95% CI 13.0 to 28.2), near the published 8 to 22 (Clemens, FanGraphs, 28 April 2026). On one plane it is 32.8 (95% CI 29.3 to 36.6). Umpires moved together, with a between-umpire spread of 0.04 (95% CI 0.00 to 0.10) inches.

Conclusion

The zone had already lost 23.4 (95% CI 18.6 to 28.3) square inches in 2025, before any challenge was made, so comparisons that start from 2025 miss that step entirely. The 2026 step is 59 percent of the two combined (95% CI 53 to 65). The break dates to 2025. The contribution is the two steps with intervals against three old-rule seasons, and the measured drift, a caveat for two-season comparisons. Its limit is the failed placebo: the design cannot assign either step to its rule. A binned cross-check misses its ± 3 square inch tolerance, finding both steps 6.0 and 7.6 larger. The sealed-set definition and acceptance criteria were committed publicly before the sealed set was opened. Games from 22 September 2026 and the postseason are held out.

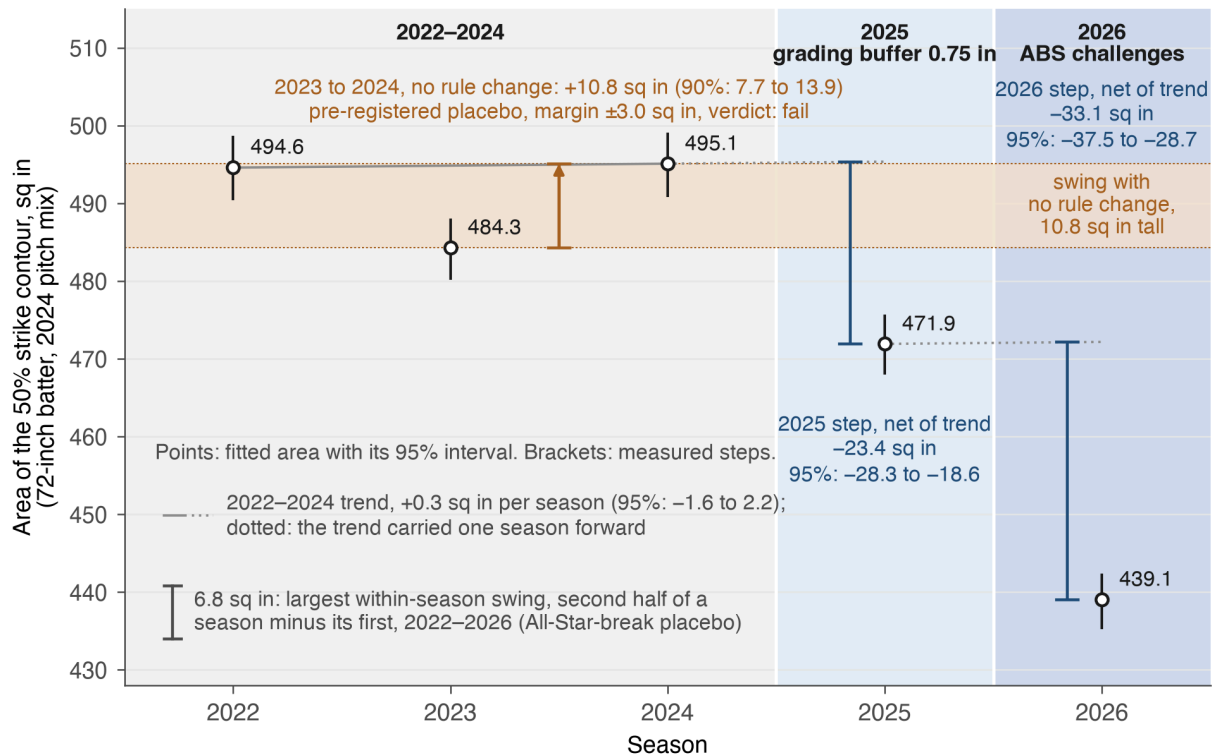


Figure 1. Area of the 50 percent strike contour, 2022–2026 (72-inch batter, 2024 pitch mix), 95% intervals. Steps net of the 2022–2024 trend: 2025, –23.4 sq in; 2026, –33.1 sq in. From 2023 to 2024, with no rule change, the area rose 10.8 sq in (90%: 7.7 to 13.9), failing the pre-registered placebo; the shaded band shows that swing.

Change in the fitted 50 percent contour	Top edge (in)	Bottom edge (in)	Half-width (in)	Area (sq in)	Shadow-band strike rate (pp)
2022-2024 trend, per season	0.13 [0.09, 0.18]	0.06 [0.02, 0.09]	-0.03 [-0.06, -0.01]	0.3 [-1.6, 2.2]	-0.09 [-0.41, 0.24]
2023 to 2024, no rule change (pre-registered placebo, 90% interval; margins ± 3 sq in and ± 0.5 pp): FAILED	—	—	—	10.8 [7.7, 13.9] 90% interval	1.54 [1.03, 2.04] 90% interval
2025 step, net of trend (grading buffer 0.75 in)	-0.18 [-0.32, -0.05]	0.18 [0.08, 0.29]	-0.41 [-0.49, -0.33]	-23.4 [-28.3, -18.6]	-4.46 [-5.28, -3.63]
2026 step, net of trend (ABS challenges)	-0.63 [-0.76, -0.50]	0.65 [0.55, 0.74]	-0.25 [-0.32, -0.16]	-33.1 [-37.5, -28.7]	-6.05 [-6.81, -5.31]
2024 to 2026, total	-0.54 [-0.65, -0.43]	0.94 [0.85, 1.03]	-0.72 [-0.80, -0.64]	-56.0 [-59.6, -52.0]	-10.69 [-11.28, -10.04]
Plate-plane component, 2025 read at mid-plate minus at the front	-0.77 [-0.94, -0.61]	-1.11 [-1.24, -0.98]	0.12 [-0.01, 0.25]	12.9 [6.3, 19.2]	—

Cells are point [95% interval] from coefficient draws of the fitted model; the placebo row prints its pre-registered 90% interval. A positive value moves an edge up or outward, or enlarges the area. Because a season pair with no rule change moved beyond its margin, no step is attributed to its rule; the 2025 and 2026 labels name the period only. Dashes: not estimated for that row.

Table 1. Measured changes in the fitted 50 percent strike contour, 72-inch batter, 2024 pitch mix, read at mid-plate. Cells are point [95% interval]; the placebo row uses its pre-registered 90% interval. The 2023-to-2024 change, with no rule change, exceeded its margin, so the 2025 and 2026 steps are labelled by period and not attributed to a rule.